Visvesvaraya Technological University
BELAGAVI, KARNATAKA.

ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ
ಬೆಳಗಾವಿ, ಕರ್ನಾಟಕ

A PROJECT PHASE-1 REPORT

ON

"Car Parking Slot Detection Using YOLO Algorithm"

Submitted to Visvesvaraya Technological University in partial fulfillment of the requirement for the award of Bachelor of Engineering degree in Computer Science and Engineering.

Submitted by

DEEPTHI B A	**4JN20CS026**
DHANYATHA B	**4JN20CS027**
TANMAYEE SHARVANI	**4JN20CS111**
VARSHITHA H G	**4JN20CS116**

Under the guidance of

Mr. Mohan H G B.E., M.Tech.
Assistant Professor, Dept. of CS&E,
JNNCE, Shivamogga.

Department of Computer Science & Engineering
Jawaharlal Nehru New College of Engineering
Shivamogga - 577 204

December 2023

National Education Society ®

Jawaharlal Nehru New College of Engineering

Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the project entitled

"Car Parking Slot Detection Using YOLO Algorithm"

Submitted by

DEEPTHI B A	**4JN20CS026**
DHANYATHA B	**4JN20CS027**
TANMAYEE SHARVANI	**4JN20CS111**
VARSHITHA H G	**4JN20CS116**

Students of 7th semester B.E. CS&E, in partial fulfillment of the requirement for the award of degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during the year 2023-24.

Signature of Guide

Mr. Mohan H G B.E., M.Tech.
Assistant Professor, Dept. of CS&E

Signature of HOD

Dr. Jalesh Kumar M.Tech, Ph.D.
Professor and Head, Dept. of CS&E

ABSTRACT

Finding free slots in the parking area has become a major problem that the modern society is facing nowadays, which also leads to traffic overcrowding at the parking area. As the number of vehicles are increasing at a quick rate find parking slots has become difficult. In traditional parking system, a person has to find vacant slot or a gate keeper needs to inform about the available parking slot at parking area. To address this issue, a system is designed, where the user will be able to know the number of vacant slots in a particular parking area along with total number of slots at the entrance of the parking area. Today's most popular object detection model YOLO, is trained to achieve desired results with good precision scores at real-time speed. The model will be getting all the parking slots that are present in the parking area, to which marking boxes are drawn around for training. Real time processing is performed on the obtained data by the model to find whether the slot is vacant or occupied. Thus, it gives the information of available vacant slots to park the car at parking area.

ACKNOWLEDGEMENT

On presenting the Machine Learning with Application Development Major Project report on "**Car Parking Slot Detection Using YOLO Algorithm**", we feel great to express our humble feelings of thanks to all those who have helped us directly or indirectly in the successful completion of the project work.

We would like to thank our respected guide **Mr. Mohan H.G**, Assistant Professor, Dept. of CSE, JNNCE, who have helped us a lot in completing this task, for their continuous encouragement and guidance throughout the project work.

We would like to thank our respected project coordinators **Dr. Poornima K.M** Professor, **Mr. Hiriyanna G.S**, **Mrs. Pushpa R.N** and **Dr. Ganavi M,** Assistant Professors, Dept. of CSE, JNNCE who have extended their warm support with respect to all aspects of our project.

We would like to thank **Dr. Jalesh Kumar**, Professor and Head, Dept. of CSE, JNNCE,Shivamogga and **Dr. Manjunatha P**, Principal, JNNCE, Shivamogga for all their support and encouragement. We are grateful to the Department of Computer Science and Engineering and our institution Jawaharlal Nehru New College of Engineering and for imparting us the knowledge with which we can do our best. Finally, we also would like to thank the whole teaching and non-teaching staff of Computer Science and Engineering Department.

Thanking you all,

Project Associates,

DEEPTHI B A	4JN20CS026
DHANYATHA B	4JN20CS027
TANMAYEE SHARVANI	4JN20CS111
VARSHITHA B	4JN20CS116

TABEL OF CONTENTS

LIST OF FIGURES